Energy Resource Innovations

Chapter 9

Big Question

What other energy resources are being developed?

You have learned about nonrenewable energy resources such as fossil fuels and nuclear power. You have also learned about renewable energy such as solar, wind, tidal, geothermal, and hydroelectric power. People are developing other resources of energy as well.

The ocean floor and the layers of sediment under Arctic permafrost contain reservoirs of oil and natural gas. They also contain deposits of a fossil fuel known as methane hydrate. It is an icelike substance made of methane and water.

People do not know the potential impacts of extracting methane hydrate from the ocean floor or Arctic permafrost. More research is required to learn how to handle, transport, and use methane hydrate.



Methane hydrate is sometimes called "fire ice" because it looks like snow but can be lit with a match.

People Are Developing Biofuels

Biofuels are made of carbon-rich chemicals that form in living things. They are produced from algae, corn plants, and sugarcane. One popular biofuel is ethanol. Ethanol is made by allowing the remains of plants, such as corn husks and stalks, to ferment, or break down in a container without oxygen. This is a type of alcohol that can be mixed with other fuels such as gasoline and burned.

Vocabulary

biofuel, n. a combustible (burnable) fuel made from the remains of recently living organisms

One benefit of using biofuels is that they are renewable. A major cost of biofuels is that they put carbon dioxide into the atmosphere and contribute to changing climate. As a result, they may not be much better for the environment than fossil fuels. They may also take up a lot of land space that could be better used for other things.



Sugarcane can be processed into ethanol, a biofuel.

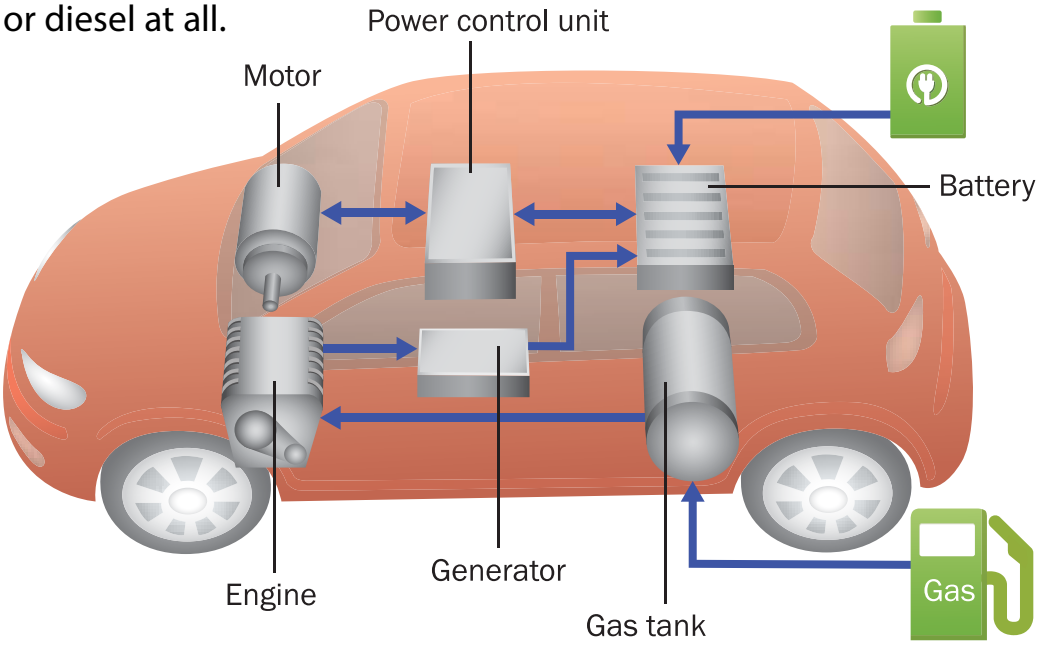
Renewable Energy for Cars

Electric automobiles and hybrid automobiles use electricity. The electric motors in these vehicles can generate electricity when the brakes are applied. The energy of motion of the car moving forward is converted to mechanical energy inside the motor. That mechanical energy then becomes electricity, which is stored in a battery. The battery supplies electricity to the motor when the driver presses the accelerator pedal to make the car move. This is called regenerative braking.

Vocabulary

hybrid, n. a combination of two types; a vehicle that uses two different sources of fuel

Cars that run on batteries either have a second engine that runs on gasoline, or they must be plugged into an electrical outlet for the battery to be charged between uses. Cars with second engines are called **hybrids**. Cars that are fully electric do not use any gasoline or diesel at all.



In a hybrid car, electricity is generated by both a gasoline-powered engine and regenerative braking technology. Some can also be plugged into an electrical outlet to be charged.

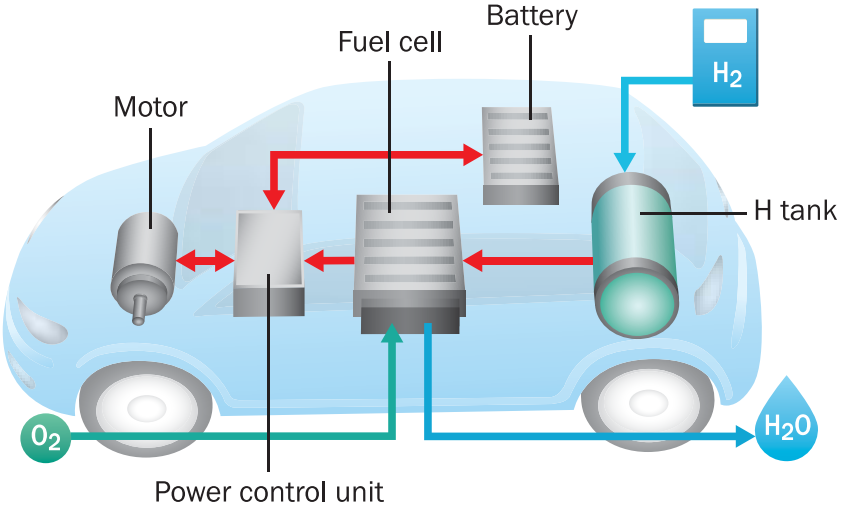
People Are Developing Fuel Cells

Another type of fuel used in some automobiles is hydrogen. Hydrogen gas can be produced by running electricity through water. The gas is captured and compressed into a tank, which is placed in a vehicle and filled at a station. When the hydrogen reacts with oxygen in the **fuel cell**, electricity is generated. This electricity powers the motor, and the motor recovers energy while braking. It stores that energy in the battery.

Hydrogen fuel cells can be used in buses, trucks, and some smaller automobiles. The energy that is used to produce and transport the hydrogen gas is often produced by burning fossil fuels. The overall effect of using hydrogen fuel cells is not necessarily a major improvement on the use of gasoline or diesel technology. There might be less pollution coming out of the vehicles but about as much pollution going into the atmosphere overall.

Vocabulary

fuel cell, n. a device in which a chemical reaction takes place to produce electricity, similar to a battery



Hydrogen gas can be pumped into a tank. The hydrogen reacts with oxygen in the fuel cell, producing electricity for the motor and the main battery. Regenerative braking helps charge the battery, too.

New Materials Can Harness Energy of Motion

Scientists hope to capture some of the energy that humans use and release through everyday activities such as walking and jogging. One example is a sidewalk made of triangular tiles that fit together in a puzzle-like pattern. Each corner of each tile has a small generator. When the tile is pressed downward as someone steps on it, the generators produce small amounts of electricity. Just one person walking on a few tiles does not produce much energy. But a lot of people walking on such a sidewalk all day can add up.

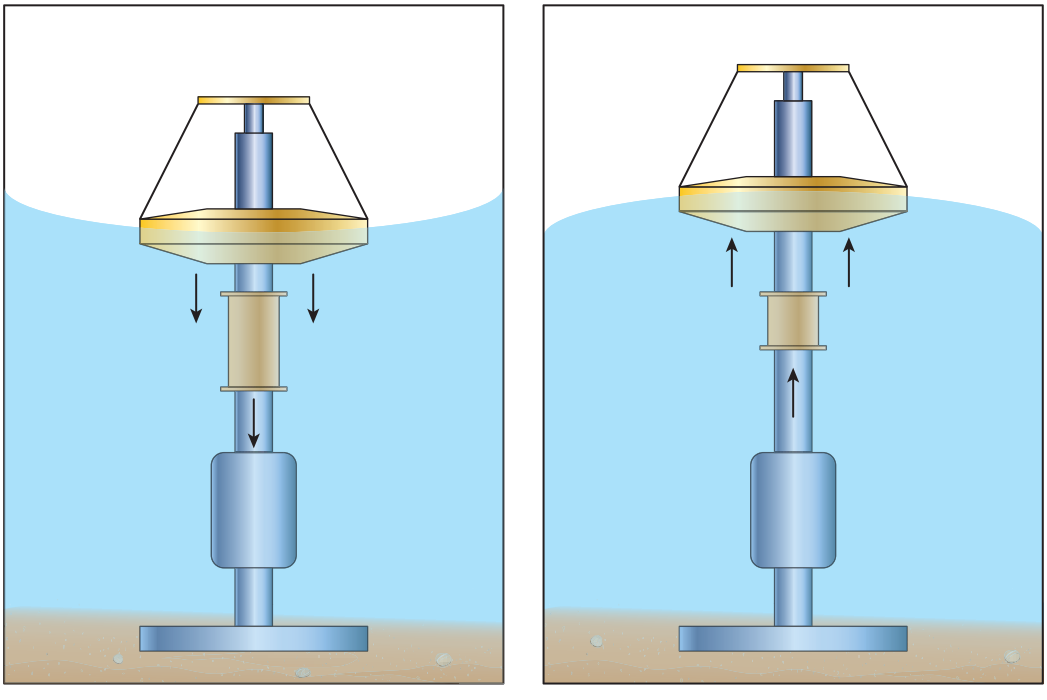
There is also technology being tested in clothing that involves special threads that interact as the clothing moves to produce small amounts of electricity. Will people one day plug their clothing into the grid to add electricity or reduce their electric bill? Probably not. But small-scale electricity production could be enough to charge a mobile phone or power another small device.



Each step on this experimental sidewalk captures energy of motion and converts it into electricity.

Energy from Waves

Engineers can harness the energy in ocean waves through different types of technology. One example is a point absorber. It is a floating device that moves up and down with the waves. It is attached to a stem anchored to the seafloor. A piston connects the pieces. As the floating part bobs up and down, the piston generates electricity. A series of point absorbers can be placed in an area with regular wave action and then connected by a cable to a grid onshore. Because waves are made by wind, this is really another way of harnessing wind energy. In this case, water delivers the motion instead of blades of a large turbine.



When the low point, or trough, of a wave passes through, the point absorber drops. When the high point, or crest, passes through, the point absorber is raised. This motion transforms mechanical energy into electricity that can be sent to the shore.